WORKED SOLUTION

STEP I 1997, Question 9

Mechanics • Relative Velocity

STEP I 1997, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q9

The key idea

From the roadside everything is moving, but from a car everything is still: the cars become a fixed row of blocks with gaps of length c, and the chicken walks across in a straight line with its *relative* velocity. It gets through only if, while it is inside the lane of width a, it drifts sideways (relative to the cars) by no more than the gap c. That single condition gives the inequality, and the minimum speed then comes from maximising $c\sin\theta + a\cos\theta$, a standard $R\cos(\theta - \alpha)$ job.

Solution

We model the chicken as a particle and the cars as rectangles moving at speed V, each occupying the same strip of width a across the road. Take the x-axis along the road in the direction of traffic and the y-axis across it, so the chicken's velocity is $(u\cos\theta,\ u\sin\theta)$ with $0 < \theta < \pi$. Only the strip containing the cars matters: elsewhere on the road the chicken is never in danger.

The inequality

Relative to the cars, the chicken's velocity is

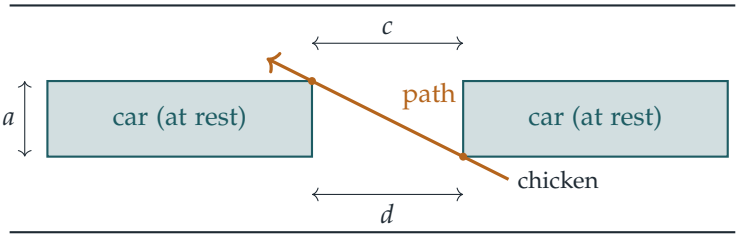
$$\mathbf{w} = (u\cos\theta - V,\ u\sin\theta).$$

In the cars' frame the cars are at rest, and the chicken moves in a straight line at constant velocity $\mathbf{w}$. It spends time

$$\tau = \frac{a}{u\sin\theta}$$

crossing the strip of width a, and during that time it slips backwards (against the traffic, relative to the cars) by

$$d = (V - u\cos\theta)\,\tau = \frac{a(V - u\cos\theta)}{u\sin\theta}.$$



The picture shows the extreme case $d = c$, where the path runs exactly along the diagonal of the gap.

Ask yourself Why must the drift d be at most c?

In the cars' frame the chicken's path through the strip covers a stretch of length $|d|$ along the road, and every point of that stretch is passed at some height inside the lane. The stationary cars leave free only gaps of length c, so the whole stretch must fit inside one gap: $|d| \le c$. (Conversely, if $|d| \le c$, the chicken can time its start so that the stretch lies inside a gap, so it crosses safely.)

In particular $d \le c$, that is,

$$a(V - u\cos\theta) \le c\,u\sin\theta, \qquad \text{so} \qquad Va \le u\,(c\sin\theta + a\cos\theta).$$

The left-hand side is positive, so $c\sin\theta + a\cos\theta > 0$ for any safe crossing, and we may divide by it:

$$u \ge \frac{Va}{c\sin\theta + a\cos\theta},$$

as required. (The other half of the condition, $d \ge -c$, stops a fast chicken from overtaking the gap and running into the car in front; it plays no part in the minimum below.)

The least speed and the crossing time

Write $c\sin\theta + a\cos\theta = \sqrt{a^2+c^2}\,\cos(\theta - \alpha)$, where α is the acute angle with

$$\cos\alpha = \frac{a}{\sqrt{a^2+c^2}}, \qquad \sin\alpha = \frac{c}{\sqrt{a^2+c^2}}, \qquad \tan\alpha = \frac{c}{a}.$$

So $c\sin\theta + a\cos\theta \le \sqrt{a^2+c^2}$, with equality only when $\theta = \alpha$. Hence for every direction

$$u \ge \frac{Va}{c\sin\theta + a\cos\theta} \ge \frac{Va}{\sqrt{a^2+c^2}},$$

and both inequalities are equalities when $\theta = \alpha$ and $u = Va/\sqrt{a^2+c^2}$. That choice really is safe: it gives $d = c$ exactly (the path of the figure, corner to corner), and $-c \le d \le c$ holds. So the least possible uniform speed is $Va/\sqrt{a^2+c^2}$, at angle $\alpha = \arctan(c/a)$.

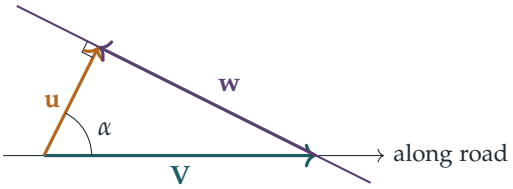
The chicken crosses the full breadth b with velocity component $u\sin\alpha$ across the road, so the time taken is

$$\frac{b}{u\sin\alpha} = b\cdot\frac{\sqrt{a^2+c^2}}{Va}\cdot\frac{\sqrt{a^2+c^2}}{c} = \frac{b\,(a^2+c^2)}{Vac} = \frac{b}{V}\left(\frac{c}{a}+\frac{a}{c}\right),$$

which is the required result.

Ask yourself Why does the least speed come with $\tan\theta = c/a$?

Look at the velocities. The chicken's velocity is $\mathbf{u} = \mathbf{V} + \mathbf{w}$, and at best the relative velocity $\mathbf{w}$ points along the diagonal of the gap, direction $(-c, a)$. So the tip of $\mathbf{u}$ lies on the line through the tip of $\mathbf{V}$ in that direction, and the shortest $\mathbf{u}$ is the perpendicular from the origin to that line. Its direction (a, c) is at angle $\arctan(c/a)$ to the road, and its length is $V\sin(\text{angle of the diagonal}) = Va/\sqrt{a^2+c^2}$.



Checks

- Units: $Va/\sqrt{a^2+c^2}$ is a speed, and $\frac{b}{V}\left(\frac{c}{a}+\frac{a}{c}\right)$ is a length over a speed, a time.
- Limits: if $\theta = \frac{\pi}{2}$ (straight across), the bound is $u \geq Va/c$: the chicken must cross the width a in the time c/V that a gap takes to pass. That is clearly right.
- Numbers: with $a = 1$, $c = 2$, $V = 2$, $b = 5$ the least speed is $2/\sqrt{5} \approx 0.894$ at $\tan\theta = 2$, and the crossing time is $\frac{5}{2}\left(2+\frac{1}{2}\right) = 6.25$. A direct simulation of cars and chicken in the road frame agrees: it finds a safe start time at 1.01 times this speed and none at 0.99 times it.

Common slips

- Using the crossing time for the whole road, $b/(u\sin\theta)$, in the safety condition. Only the time spent in the lane of width a matters.
- Getting the sign of the relative velocity wrong: relative to the cars the chicken moves at $u\cos\theta - V$ along the road, not $u\cos\theta + V$.

Beyond STEP

Changing to the frame in which the obstacles are at rest is a Galilean transformation, and choosing the frame in which a problem is static is one of the most useful habits in

mechanics. The minimisation is also a small optimisation problem: the shortest vector from a point to a line is the perpendicular, which you will meet again in linear algebra as orthogonal projection. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems on relative motion in this spirit.

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